Maths Advanced Calculus

Engineering Technology

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Maths Advanced Calculus (A11923)

Short Title:	Maths Advanced Calculus		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Mathematics and Physics		
Author	CLAIRE.FITZPATRICK		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

This module presents a standard course in multivariable calculus (including vector analysis) together with (i) an introduction to the representation of periodic functions with trigonometric series and (ii) a brief introduction to the mathematical analysis of random variables.

Programmes

`MTHE-0031`	BEng (Hons) in Electrical Engineering (`WD_ETRIC_B`)	2 / 4 / M
	BEng (Hons) in Electrical and Automation Engineering (International) (`WD_ETRIC_BI`)	2 / 4 / M
`MTHE-0031`	BEng (Hons) in Electronic Engineering (`WD_EONIC_B`)	2 / 4 / M
	BEng (Hons) in Information Engineering (International) (`WD_EEELC_BI`)	2 / 4 / M
`MTHE-0031`	BEng (Hons) (`WD_ETRIC_B`)	2 / 4 / M
`MTHE-0031`	BSc (Hons) in Physics for Modern Technology (`WD_KPHTE_B`)	2 / 4 / M

Indicative Content

- Multivariable differential calculus; chain rules; directional derivatives; gradient function; differential operators: div, curl and their composition; Taylor expansion; critical points; classification using function value, first and second derivative tests; tangent plane; optimisation
- Multivariable integral calculus; line Integrals; iterated integrals; surface integrals; Greens theorem; Stokes theorem, divergence theorem
- Fourier Series; comparison of Taylor and Fourier approximation of functions. Use of a table of Fourier series; derivation from first principles. properties of Fourier series approximation
- Laws of probability; conditional probability; discrete and continuous probability distributions

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Analyse and manipulate expressions involving differential operators.
2. Compute iterated integrals, line integrals and surface integrals.
3. Use the integral theorems (Divergence, Stokes and Greens).
4. Describe the criterion on which the approximation of real periodic functions using trigonometric series is based and to derive an approximation from first principles and to discuss its properties.
5. Analyse change in a mathematical model when a deterministic variable is replaced by a random variable and to calculate moments for a simple distribution.
6. Use standard mathematical software/algorithms to demonstrate the application of mathematical concepts to engineering problems.

Learning and Teaching Methods

- Delivery of the subject will be through a mixture of lectures and directed problem-solving sessions.
- The lectures will develop theory, lead students through worked examples and introduce the context for the subject material.
- The tutorial classes will underpin and rehearse the skills demonstrated in the lectures. Students will be required to construct valid and precise mathematical arguments.
- The laboratory sessions consist of a number of computer based practicals to reinforce the concepts covered during the lectures using mathematical software such as OCTAVE/MATLAB/Python.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3,4,5
Continuous Assessment	30%	
Practical	15%	6
In-Class Assessment	15%	1,2,3,4,5

Assessment Criteria

<40%: Has not attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter or demonstrated a sufficient level of the skills expected on completion of this module.

40%-49%: Has attained the module learning outcomes at a basic level, demonstrated a basic knowledge of the associated subject matter, and achieved a basic level of the skills required for the subject matter.

50%-59%: Has attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

60%-69%: Has attained the module learning outcomes at a very good level, demonstrated a detailed knowledge of the associated subject matter, and achieved a very good level of the skills required for the subject matter.

70%-100%: Has attained the module learning outcomes at an excellent level, demonstrated a comprehensive knowledge of the associated subject matter, achieved an excellent level of the skills required for the subject matter, and demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Tutorial	12	
Practical		
Independent Learning	87	

Supplementary Material

- Greenberg, M.D. _Advanced Engineering Mathematics_. .USA: Prentice Hall, .2006.
- McCallum, W.G. and D. Hughes-Hallett. _Multivariable Calculus_. .USA: Wiley, .2012.

Requested Resources

- Room Type: Computer Lab